Lecture Notes on

Black Holes

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These notes expand on our Astrophysics Club presentation on black holes. We go slightly beyond the slides to include short derivations and deeper explanations, while keeping the discussion accessible.

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1 What Is a Black Hole?

- A black hole is a region of spacetime where gravity is so strong that nothing — not even light — can escape.
- It is not a "hole" in space, but a region of extreme curvature.
- The defining feature is the existence of an **event horizon**.

Key idea:

- In general relativity, gravity is not a force.
- Mass and energy curve spacetime.
- Objects follow the geometry of spacetime.

Near a black hole, curvature becomes so extreme that all possible future paths point inward.

2 Formation of Black Holes

2.1 Supernova Collapse

- Massive stars ($\gtrsim 25M_\odot$) fuse heavier and heavier elements.
- Once iron forms in the core, fusion can no longer release energy.
- The outward radiation pressure disappears.
- Gravity causes catastrophic collapse.
- The outer layers explode (supernova).
- If the remaining core is sufficiently massive, collapse continues into a black hole.

Think of it as concentrating stellar mass into an extremely small radius.

3 The Schwarzschild Radius

If a mass M is compressed inside a critical radius, a black hole forms.
This radius is

$$R_s = \frac{2GM}{c^2}.$$

3.1 Derivation via Escape Velocity

Start with escape velocity:

$$v_{\text{esc}} = \sqrt{\frac{2GM}{R}}.$$

A black hole forms when even light cannot escape:

$$v_{\text{esc}} = c.$$

Setting them equal:

$$c^2 = \frac{2GM}{R}.$$

Solving:

$$R = \frac{2GM}{c^2}.$$

This classical derivation surprisingly gives the exact Schwarzschild radius.

3.2 Example: Earth

$$R_s^{(\text{Earth})} \approx 0.35 \text{ inches}.$$

If Earth's mass were compressed into a sphere about 0.7 inches across, it would become a black hole.

4 Why Light Cannot Escape

- Escape velocity increases as radius decreases.
- At $R = R_s$, escape velocity equals c.
- Inside this radius, all future-directed paths lead inward.
- Light cannot increase its speed beyond c, so it cannot escape.

This boundary is the event horizon.

5 The Event Horizon

- The event horizon is the boundary surrounding a black hole.
- It is not a physical surface.
- It marks the point of no return.
- To distant observers:
  - Infalling objects appear to slow down.
  - They appear to freeze at the horizon.
  - They become increasingly redshifted.

The black region we observe is the shadow created by this geometry.

6 Observational Evidence

- First direct image: 2019.
- Black hole in M87 galaxy.
- Distance: ~ 55 million light-years.
- Observed by the Event Horizon Telescope.
- The telescope is a global array of radio telescopes functioning as an Earth-sized interferometer.

We have also detected gravitational waves from black hole mergers.

7 Relativistic Effects Near Black Holes

7.1 Time Dilation

- Clocks run slower in stronger gravitational fields.
- Near a black hole, time dilation becomes extreme.
- To distant observers, time appears to stop at the horizon.

This effect explains the dramatic time difference depicted in *Interstellar*.

7.2 Spaghettification

- Gravity decreases with distance.
- The difference in gravitational force across an object is called a tidal force.
- Near a black hole, tidal forces can become enormous.
- Objects stretch into long thin shapes.

8 Death of Black Holes: Hawking Radiation

Black holes are not completely eternal.

8.1 Quantum Vacuum Fluctuations

- Quantum field theory predicts that virtual particle pairs constantly form.
- Normally they annihilate almost immediately.
- Near the event horizon, one particle can fall in while the other escapes.

8.2 Energy Accounting

- The escaping particle becomes real and carries away energy.
- By conservation of energy, the infalling particle must have negative energy (relative to infinity).
- Because $E = mc^2$, removing energy reduces mass.
- The black hole slowly evaporates.

This process is called **Hawking radiation**.

8.3 Timescale

The evaporation time scales approximately as

$$t \propto M^3.$$

Supermassive black holes may take

$$\sim 10^{100} \text{ years}$$

to fully evaporate.